

Tutorial No-6

1] Write a short note on lean software development

Principles of lean software development -
eliminate waste -

Remove anything that doesn't add value.
Regular reviews of bottlenecks and inefficiencies

Build in Quality -

ensure quality through practices like
pair programming and test driven development.

Amplify learning -

Promote continuous learning and knowledge sharing

Delay commitment

Make irreversible decision as late as possible to stay flexible and adaptive to change.

Deliver fast -

- Release products quickly, gather feedback early, iterate based on insights.

6. Respect people -

- Foster a collaborative and respectful work environment.

7. Optimize the whole -

- Improve the entire process end-to-end rather than focusing only on individual parts.

Q.27 Write short note on Scrum.



- Scrum is a lightweight framework for managing complex projects, particularly in software development.
- Designed to help Teams work collaboratively and efficiently.

Core Concept -

- Scrum emphasizes iterative development, breaking down complex projects into short focused cycles called sprints.

Key elements -

① Sprint - Time boxed periods during which specific work is completed.

② Roles -

- Product owner - defines the what (Product backlog)
- Scrum master - facilitates the scrum process
- development team - delivers the work.

③ Events -

- Sprint planning - defining the work for the sprint
- Daily scrum - A short daily meeting for progress updates.
- Sprint review - demonstrating the completed work.

④ Artifacts -

- Product backlog - A prioritized list of features
- Sprint Backlog - the work selected for specific sprint.
- Increment - the usable outcome of each sprint.

⑤ Purpose -

- To enhance teamwork, improve communication and deliver valuable products or services in flexible and responsive manner.

Q.3] Explain extreme programming?

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Extreme programming Advantages-

- extreme programming solves the following problems often.
- faced in software development project-

① Slipped schedules-

- achievable development cycles ensure timely deliveries

② Cancelled Project-

- focus on continuous customer involvement ensures transparency with the customer

③ Costs incurred in changes

④ Production and post-delivering defects

⑤ Misunderstanding the business and domain

⑥ Business changes

⑦ staff turnover

* extreme programming values-

① Communication

② Simplicity

③ Feedback

④ Courage

⑤ Respect

Q.4) explain different design and development practices in Agile project

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① Prioritize user experience (UX)

- Agile Teams recognize the importance of user interface and UX as the system itself, so design considerations are woven into the development process.

② Iterative Design

- Design is not one-time task but a continuous process evolving alongside the development process.

③ Simplicity-

- Teams strive to design solutions that are as simple as possible while still meeting user needs.

④ Documentation-

- Agile design encourages documentation especially for complex aspects of design but avoids over documentation.

⑤ Agile Data modeling-

- For database design, evaluating database modeling techniques are employed, including agile data modeling database refactoring.

Development practices

① Continuous integration

- Agile team use continuous integration where code changes are frequently integrated into shared repository.

② Test driven development

- develops write tests before writing code.

3. Refactoring -

Agile team regularly refactors their code to improve its structure

4. Pair programming

Developers work in pairs, sharing screen and keyboard, code review.

5. explain Agile Manifesto and principles?

Agile manifesto is a document outlining core values and principles for agile software development, focusing on flexibility, collaboration and customer satisfaction

It emphasises individuals and interaction over processes and tools, working software over comprehensive documentation, customer collaboration over contract negotiation and responding to change over following plan.

Agile manifesto core values:-

- ① Individuals and interaction over processes and tools
- ② Working software over comprehensive documentation.
- ③ Customer Collaboration over contract negotiation
- ④ Responding to change over following a plan

#12 principles behind the Agile manifesto-

- ① Customer satisfaction
- ② Changing requirements
- ③ Frequent Delivery
- ④ Regular Communication
- ⑤ Motivated individuals
- ⑥ face-to-face meetings
- ⑦ Measure outputs

Sustainable development processes

Technical excellence and good design

Simplicity

Self-sufficient teams

Continuous improvement.

CGK